



2/9/2023

Christopher Andujar
398 Catalina Dr
Mooresville, NC 28117

Re: 398 Catalina Dr; Mooresville, NC
Date of Observation: 2/9/2023

Christopher:

I am providing you this letter in reference to your request for a foundation observation of the single family home residence located at the above noted address. Prior to the observation, you discussed with me that concerns had been raised that warranted review by a licensed, structural engineer. The scope of this observation was limited to evaluating the items noted below and making repair recommendations if warranted. The findings of this report are based on visual observations only. Please note this report does not express or imply any warranty of the structure but only addresses the condition of the portion which was readily accessible and observable at the time of observation.

Observations:

I visited the property at the above noted date and time and observed the following (all location references are as viewing the property from the front):

- Iredell County property records indicate the Property is a conventionally framed house supported by a combination basement wall and crawlspace foundation system. The date of initial construction is reported to be 1995.
- The home was previously inspected by Alex Parr (dated 2/2/2023). The scope of this report is limited to addressing the structural concerns raised by the inspector noted below.
- The foundation along the right wall was constructed of multi-wythe brick masonry. The front portions of the wall contain 2-wythes of brick masonry with the exterior band blocked down to the wall at varying intervals with 2x4s. The rear section of wall contains 1-wythe of brick masonry with the exterior band frespanning 6'-4" between supports.

Conclusions & Recommendations:

The following are recommendations based on the above noted observations. Please note the recommendation below does not cover the means and methods to install the repairs. Means and methods are the responsibility of the licensed contractor performing the work. Changes/alterations to the recommendations below cannot be made without the prior written consent of the Engineer of Record (EOR). The authority having jurisdiction (AHJ) should be contacted to determine if building permits are warranted for the repairs noted below. Special Inspections are not required for this project. We recommend our firm be retained at the completion of the repairs to observe the repairs and determine if they are in general accordance with the recommendations noted below.

- The portions of wall containing 2-wythes of masonry require blocking be installed between the top of wall/band such that the maximum free span between blocking is 5'-0" or less.

The rear section of wall containing 1-wythe of masonry requires a new pier(s) be installed where the band is spanning more than 5'-0". The new piers shall be constructed of 8" CMU block per the requirements of the NCRC and bear on 20" square x 10" thick footings made of 3,000 psi concrete (28-day compressive strength). Alternatively, steel piers approved for use by the IRC/ICC may be substituted in lieu of masonry piers.

The opinions stated in this report are based on limited visual observations only. No physical testing was performed, and no calculations have been made to determine the adequacy of the structural system or its compliance with adopted building code requirements. This report was prepared for the exclusive use of the client and was not intended for any other purpose. Our report was based on the information available to us at this time. Should additional information become available, we reserve the right to determine the impact, if any, the new information may have on our opinions and conclusions and to revise our opinions and conclusions if necessary and warranted.

Architectural, mechanical, electrical, or plumbing conditions are not included, and no warranty expressed or implied as to the condition of the structure, is intended. The maximum liability of the Engineer for loss suffered by the client due to any cause is limited to the amount of the Engineer's fee paid by the client. This shall be the sole exclusive remedy for any loss suffered by the client arising out of the Engineer's performance under this agreement.

It has been a pleasure serving you. Please contact us if you have any questions or comments.

Sincerely,

Jon Vandergrift, PE
Structural Engineer

